

Nicolò Carpentieri

SYSTEMS & COMPUTER ARCHITECTURE

PhD Candidate, Systems Research Group
Technical University of Munich

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01 RESEARCH INTERESTS

My interests lie in **computer architecture and systems**, particularly **memory disaggregation** at rack scale and the scale-up interconnects, like CXL and related fabrics, that make it possible. Within that space, I focus on **cache coherence protocols** for heterogeneous accelerators, including GPUs, and on automating the design and verification of the controllers that implement them.

02 EDUCATION

Ph.D. in Computer Science

Oct 2024 -- present

Technical University of Munich

Advisor: Prof. Dr. Pramod Bhatotia

M.Sc. in Computer Engineering

2021 -- 2024

Politecnico di Torino

Thesis: Designing and Evaluating Mapping of CNN Layers on an Edge-CGRA, in collaboration with the Embedded Systems Laboratory (ESL), EPFL

B.Sc. in Electronic Engineering

2018 -- 2021

Politecnico di Torino

03 RESEARCH EXPERIENCE

Visiting Student, Embedded Systems Laboratory (ESL), EPFL

2023 -- 2024

Mapping of convolutional neural network algorithms onto CGRA architectures, in collaboration with DAUIN, Politecnico di Torino.

04 PUBLICATIONS

- [1] Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers
Nicolò Carpentieri, Anatole Lefort, David Schall, Julian Pritzi, Pramod Bhatotia
IEEE/ACM MICRO 2026
- [2] vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures
Anatole Lefort, Julian Pritzi, **Nicolò Carpentieri**, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia
ACM ASPLOS 2026 ★ **Best Paper Award**
- [3] C3: CXL Coherence Controllers for Heterogeneous Architectures
Anatole Lefort, David Schall, **Nicolò Carpentieri**, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia
IEEE HPCA 2026
- [4] Performance Evaluation of Acceleration of Convolutional Layers on OpenEdgeCGRA
Nicolò Carpentieri, Juan Saprizo, Davide Schiavone, Daniele Jahier Pagliari, David Atienza, Maurizio Martina, Alessio Burrello
ACM CF 2024

05 TALKS

ProtoSLICC: Automated Synthesis of Gem5-based Cache Coherence Controllers
gem5 Workshop @ ISCA 2025, Tokyo, Japan

Jun 2025

06 ACADEMIC SERVICE

Artifact Evaluation Committee
ACM ASPLOS 2026

2026

07 HONORS AND AWARDS

Induction in IEEE-HKN (Mu Nu Chapter), Politecnico di Torino chapter

Jun 2022

08 TEACHING

Introduction to Software Engineering (EIST) — *Examiner*
Systems Programming — *Course Organizer*

2024--2026

2026--2027

09 SUPERVISED THESES

Disaggregated, Vendor-Generic APUs with CXL
Victor Fritz Trost — B.Sc. Thesis

SS 2025

10 TECHNICAL SKILLS

C · C++ · Python · VHDL · Verilog